Mohammad Ali Bazrafshani

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CAREER PROFILE

Mechanical design researcher capable of working both independently and as part of a multidisciplinary team. Extensive experience in humanoid robotics, mechanical system integration, CAD-driven product development, actuator/mechanism design, manufacturing, assembly, testing, and multidisciplinary collaboration in academic and industrial environments. Experienced in supporting robotic systems from concept through prototyping, troubleshooting, demonstrations, and iterative hardware improvement. Motivated by complex engineering challenges and advanced problem-solving.

RELEVANT SKILLS

Mechanical Design & Manufacturing

- Design of complex assemblies and innovative robotic mechanisms (e.g., custom actuator and mechanism integration for the SURENA humanoid robot).
- Mechanical prototyping, manufacturing preparation, CAM, and assembly for multiple robotics research projects.
- Debugging and iterative redesign of mechanical subsystems to improve reliability and test readiness, including integration of additional sensors for error tracking.
- Application of cutting-edge design methods including topology optimisation and analysis-driven mechanical design.
- Advanced SolidWorks user; delivered SolidWorks training courses and workshops for students and research teams.

Electronics & Embedded Integration

- PCB design using Altium Designer, including controller boards and sensor-interface boards for robotic systems.
- Integration of embedded electronics within robotic mechanisms and compact electromechanical assemblies.
- Familiarity with electrical noise mitigation, wiring integration, and advanced sensor interfacing.

Software & Tools

- ROS, ROS2, Python, C++ and Git (basic working familiarity)

Communication

- Collaborating with multidisciplinary research teams (electronics, control, and AI) to align mechanical designs with system-level requirements.
- Delivering workshops on mechanical design improvement and CAM to support fabrication projects and student research.
- Co-author of publications in international robotics conferences (e.g., IEEE-RAS Humanoids).

Interpersonal skills

- Research collaboration with students and engineers to improve mechanical design quality and real-world applicability of robotic systems.
- Teamwork, active listening, and conflict resolution in academic and industrial environments.

Research and Analysis

- Contributed to a wide range of robotics projects at the CAST Laboratory (University of Tehran) including humanoid robots, bio inspired robots (robotic fish), flying robots (fixed wing and flapping wing), and assistive devices such as prosthetic hands, ankle and knee joints, and exoskeletons.
- Applied specialized assembly techniques including precision fits, adhesives with specific material properties, and compact sensor integration.
- Performed repeated system repair, redesign, sensor integration, and performance improvements across multiple robotic platforms.
- Designed custom actuators for the SURENA IV humanoid robot using frameless RoboDrive servomotors, harmonic drives, and integrated sensors to meet joint level torque, speed, weight, and geometric constraints.

Project Management

- Mechanical Design Lead for the SURENA IV humanoid robot, responsible for coordinating mechanical design, analysis, manufacturing, and assembly activities. (SURENA IV was recognised by ASME as a notable academic humanoid robot in 2020).
- Organized and managed multiple robot unveiling and demonstration events for industrial and external stakeholders.
- Founder of the Pooyesh Machine robotics team, which received the Best Industrial Working Group in Iran award from the Ministry of Labour (2025).

WORK EXPERIENCE

Mechanical Design Researcher, **2015-Present**
Department of Mechanical Engineering, University of Tehran, Iran

- Led mechanical design, analysis, and manufacturing activities for robotics research projects including humanoid, bio-inspired, and aerial robots.

Mechanical Design Engineer, **2019-Present**
Pooyesh Darou Pharmaceuticals Factory, Tehran, Iran

- Founded and led the design and manufacturing of robotic and automated pharmaceutical machinery.

EDUCATION

M.Sc. in Mechanical Engineering, University of Tehran **2019-2023**
Thesis: Development of adaptive friction based on lubrication phase change and its implementation in robotic, Grade: Excellent

B.Sc. in Mechanical Engineering, Shahid Bahonar University of Kerman **2009-2015**

RELEVANT SKILLS & COURSES

- Delivered SolidWorks training courses through the Mechanical Engineering Scientific Association at the University of Tehran.
- Participated in robotic system demonstrations for industrial visitors and university showcase events.
- Use of Microsoft Word, LaTeX, and Excel for technical documentation and data analysis.
- Driving license (eligible to obtain a UK driving license if required).

INTERESTS

- Volunteering, Nature walk and hiking.

REFEREES

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